

funTDA — Part A: functional network baseline (H3N2)

Persistence landscapes, FPCA, and functional permutation tests over the 17 GRNs

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Data

This analysis starts from 17 gene regulatory networks (GRNs) built from the transcriptomic response to an experimental H3N2 influenza infection (Higgins, Wu & Carey). Of the 17 subjects, 8 remained **asymptomatic** and 9 developed **symptoms**. Each network is a 250×250 adjacency matrix, weighted and **undirected** (symmetric).

i Structural check of the data

- Number of adjacency matrices found: **17** (8 asymptomatic + 9 symptomatic).
- Dimension of all matrices: **250 × 250** — OK.
- Maximum asymmetry $\max(|A - A^T|)$ over the 17 matrices: **0** (0 = perfectly symmetric / undirected).

Before turning to persistent homology, it is worth looking at the networks “by eye”. The figure below compares, for one asymptomatic and one symptomatic subject, a 60-node subgraph keeping only the highest-weight edges (threshold: 85th percentile of edge weights, same as in `make_figs.py`):

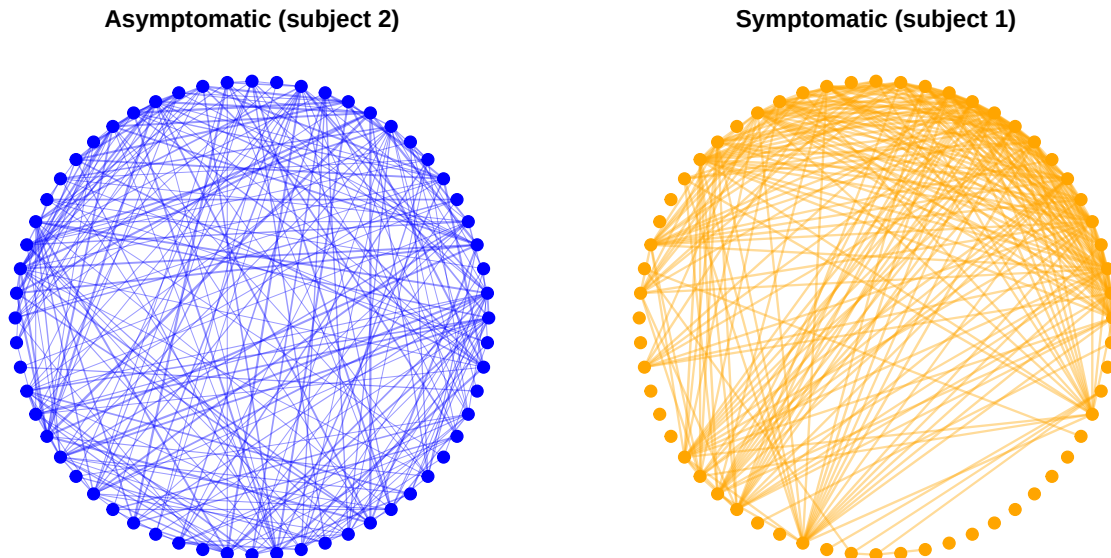


Figure 1: 60-node subgraphs (edges > 85 th weight percentile) of one asymptomatic subject (subject 2) and one symptomatic subject (subject 1). No obvious structure visually distinguishes the two clinical classes — this motivates turning to persistent homology.

No structural difference stands out between the two panels: density, node degree, and overall graph appearance are comparable despite belonging to different clinical classes. This visual indistinguishability is precisely what motivates describing each network through a topological invariant (persistent homology) rather than direct graph statistics.

Persistence diagrams for each network (previously computed in Python with `giotto-tda`, see `funTDA.ipynb`) are precomputed in `output/PH/*.csv`, with columns `Birth`, `Death`, `dimension` and no header.

Inspecting the `dimension` column of the 17 persistence CSVs confirms that the only dimensions present are `0`, `1` (H_0 connected components and H_1 loops). The baseline computes only H_0 and H_1 (the `giotto-tda` default, `homology_dimensions=(0,1)`), following `funTDA`; the analysis below therefore uses H_1 landscapes (order $r=1$, first layer $KK=1$), replicating `funTDA.R`. The H_2 / voids question is taken up with the image data, not the network baseline.

Pipeline: from persistence diagrams to functional landscapes

The `landscape(..., KK = 1)` reimplementation was validated by comparing it bit-for-bit against `TDA::landscape` over the 17 subjects (maximum absolute difference = 0; see reproducible appendix). Curves are smoothed with an order-2 B-spline basis (`nbasis = 499`) over `tseq = seq(0, 1, length.out = 500)`, identical to `funTDA.R`.

Example: persistence diagram and landscape of a representative subject

To illustrate the step from “persistence diagram” to “functional landscape”, a single representative subject is used — Symptomatic 1, the same one used in `make_figs.py` for `network_to_landscape.png` — and both representations are shown side by side, using the same `r_landscape1()` defined above.

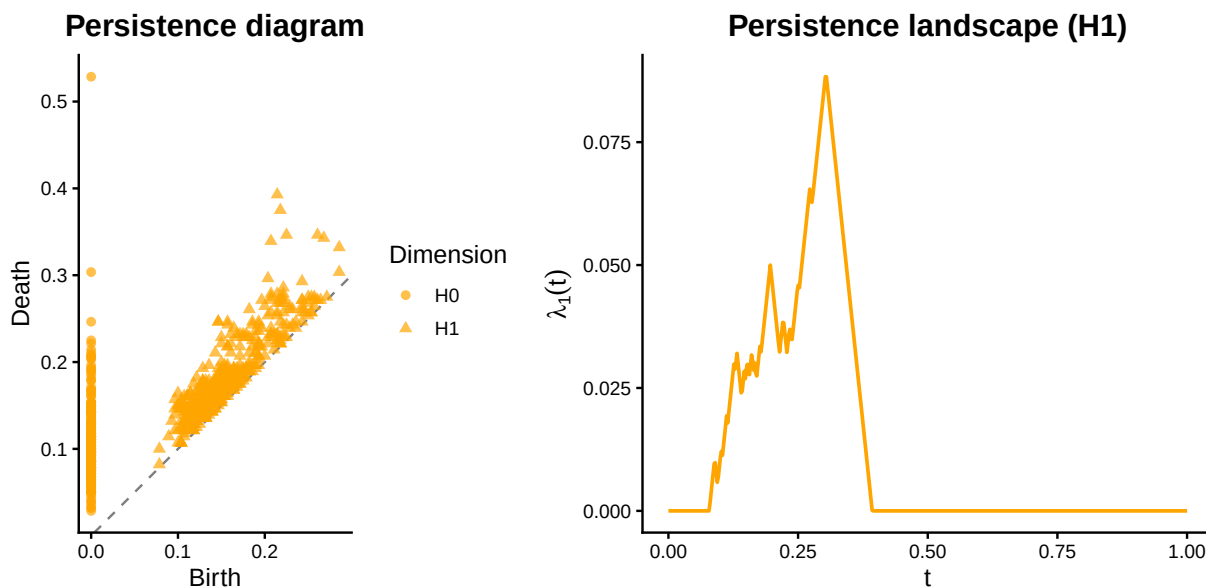


Figure 2: Representative subject (Symptomatic 1): H0/H1 persistence diagram (left) and H1 landscape $\lambda_1(t)$ (right).

Landscapes by group

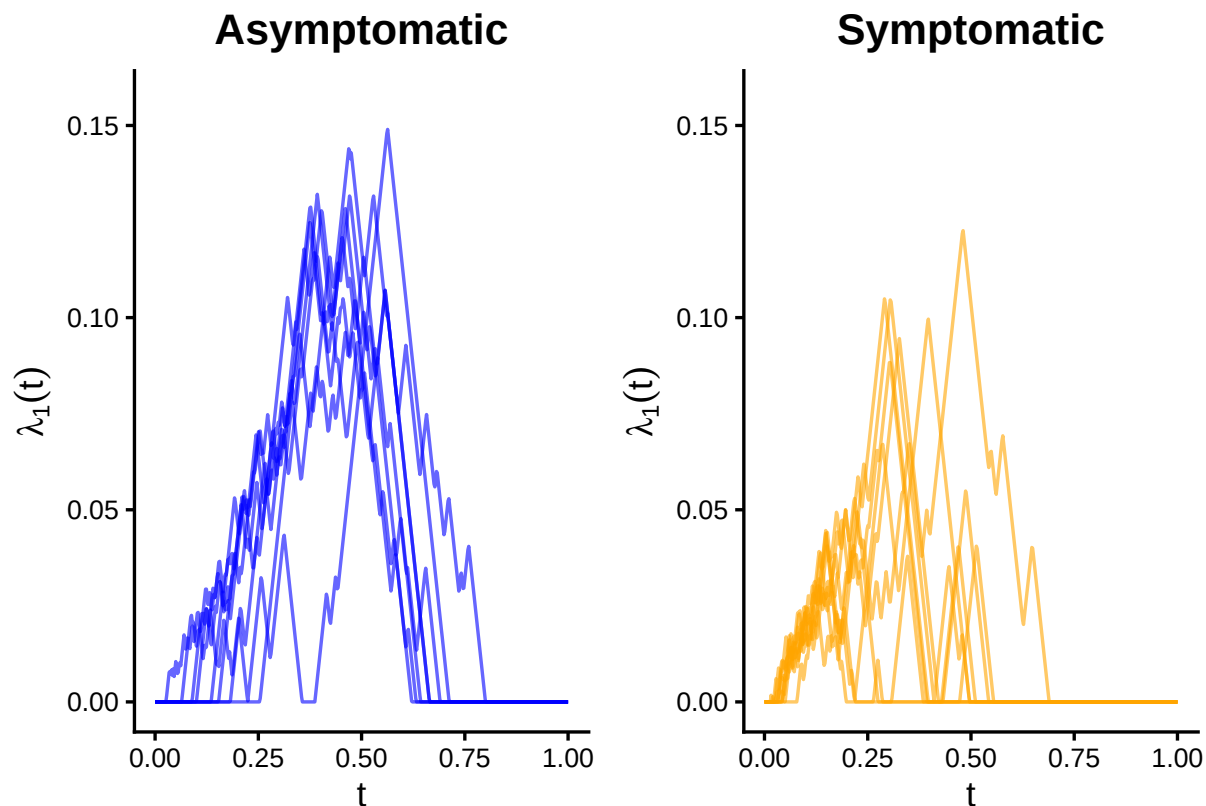


Figure 3: Individual H1 landscapes, one panel per group (same vertical scale). Asymptomatic subjects consistently occupy a band of higher peaks (~ 0.11 – 0.15), while symptomatic subjects are more variable and generally lower (~ 0.02 – 0.12) — a pattern consistent with the PC1 separation and with the permutation test ($p = 0.0011$).

Functional Principal Component Analysis (FPCA)

Table 1: Variance explained by the first two functional components

Component	% variance explained
PC1	63.54%
PC2	19.99%
Cumulative (PC1+PC2)	83.53%

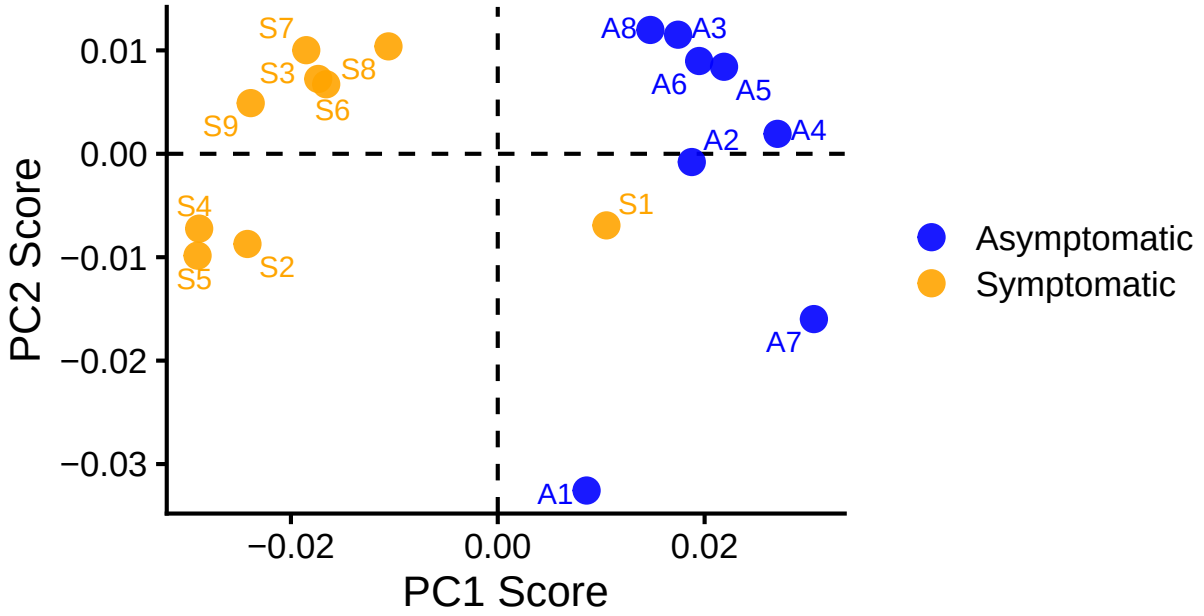


Figure 4: PC1 vs PC2 scores (FPCA over H1 landscapes), colored by group

i Full FPC scores (all 17 subjects)

ID	Group	PC1	PC2
A1	Asymptomatic	0.0086	-0.0326
A2	Asymptomatic	0.0188	-0.0008
A3	Asymptomatic	0.0174	0.0115
A4	Asymptomatic	0.0271	0.0019
A5	Asymptomatic	0.0219	0.0084
A6	Asymptomatic	0.0195	0.0090
A7	Asymptomatic	0.0306	-0.0160
A8	Asymptomatic	0.0148	0.0120
S1	Symptomatic	0.0105	-0.0069
S2	Symptomatic	-0.0242	-0.0087
S3	Symptomatic	-0.0174	0.0072
S4	Symptomatic	-0.0289	-0.0072
S5	Symptomatic	-0.0290	-0.0098
S6	Symptomatic	-0.0166	0.0067
S7	Symptomatic	-0.0185	0.0100

S8	Symptomatic	-0.0106	0.0104
S9	Symptomatic	-0.0239	0.0049

i Acceptance criteria — FPCA (baseline vs. recomputed)

Metric	Reference	Recomputed
PC1	63.5%	63.54%
PC2	20.0%	19.99%
Cumulative	83.5%	83.53%
Asymptomatic with $PC1 > 0$	8 / 8	8 / 8
Symptomatic with $PC1 < 0$	8 / 9	8 / 9

The first two functional components explain **83.5%** of the variability among landscapes ($PC1 \approx 63.5\%$, $PC2 \approx 20.0\%$). In the $PC1$ – $PC2$ plane, all 8 asymptomatic subjects fall in the $PC1 > 0$ half-plane, while 8 of the 9 symptomatic subjects fall in $PC1 < 0$: **one of the symptomatic subjects** (labeled here in a purely positional way — the way `DataAsymp/DataSymp` are built in `funTDA.R` prepends (`cbind(x1, Data)`) each new network to those already accumulated, reversing the order within each group relative to the file ID, so the “S1..S9” label should not be read as the subject’s real ID without separately verifying it) overlaps with the asymptomatic region. $PC1$ separates the two groups reasonably well; $PC2$ shows no clear separation.

Functional permutation test: asymptomatic vs. symptomatic

To avoid relying on `tperm.fd` with `nperm = 10000` (~40 s per test, infeasible for the 196 split-half tests), a vectorized version of the **same test** is used: it evaluates each functional object on the same grid `tperm.fd` uses by default (`argvals = seq(0, 1, length.out = 101)`) and permutes columns of a matrix instead of re-evaluating `fd` on every permutation. It was validated to reproduce `tperm.fd` **bit-for-bit** with the same seed (see appendix).

! Asymptomatic vs. Symptomatic test result

- **p-value** = 0.0011 (reference: 0.0011)
- **Observed $T_{\max}$** = 6.88 (reference: 6.88)
- **Critical value (95%)** = 3.72 (reference: 3.72)

With $T_{\max} = 6.88$ well above the 95% critical value (3.72) and a p-value of 0.0011, the null hypothesis of functional equality between the mean H1 landscapes of asymptomatic and symptomatic subjects is rejected: **network topologies differ statistically significantly between the two groups**.

Split-half validation (test calibration)

If the permutation test is well calibrated, randomly splitting **the same group** into two halves and comparing them against each other should yield p-values approximately uniformly distributed on

$[0, 1]$ — that is, equality should not be rejected more often than the nominal level (5%). This is checked exhaustively by considering **all** possible partitions of each group:

Table 2: Summary of the split-half validation by group

Group	mean p-value	sd(p-value)	% splits with $p < 0.05$
Symptomatic (126 splits, 4 vs 5)	0.498	0.290	5.6%
Asymptomatic (70 splits, 4 vs 4)	0.490	0.291	5.7%

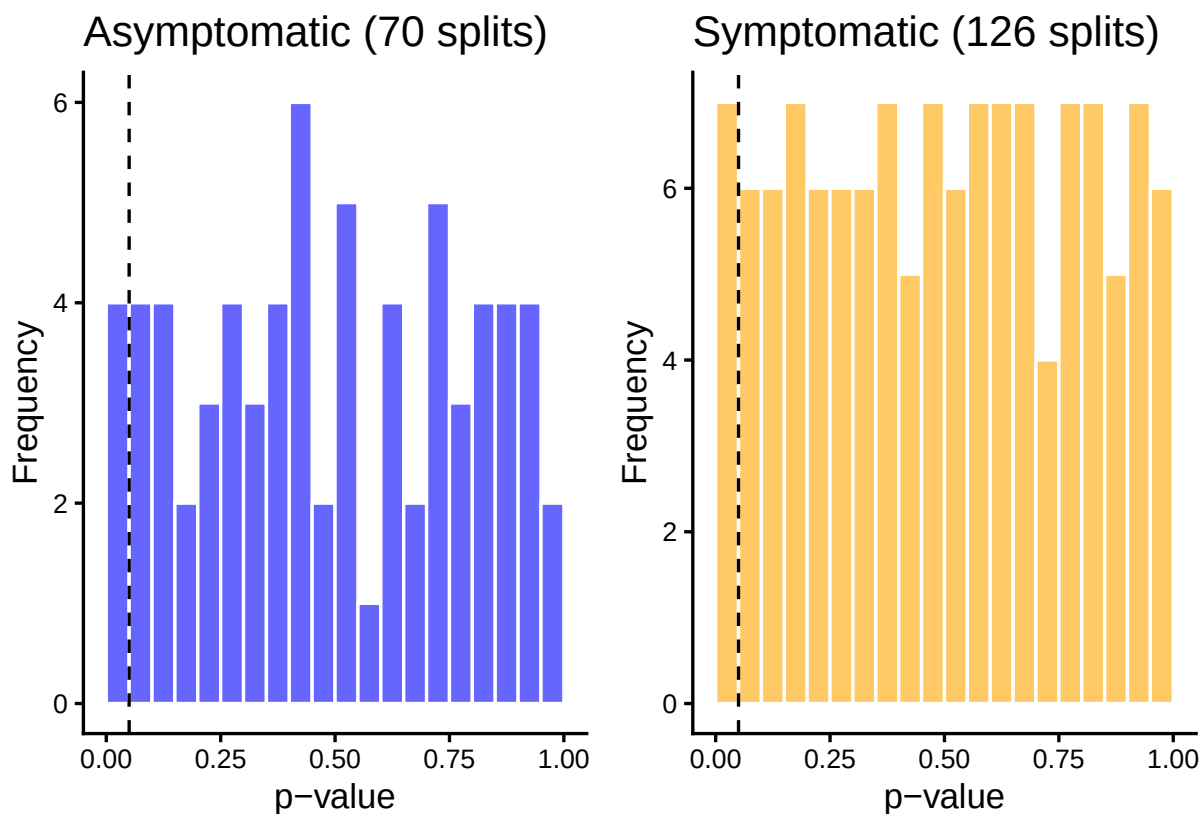


Figure 5: Distribution of split-half p-values, by group (dashed line at $p = 0.05$)

i Acceptance criteria — split-half (baseline vs. recomputed)

Metric	Reference	Recomputed
Symptomatic — mean p	0.496	0.498
Symptomatic — sd p	0.290	0.290
Symptomatic — % $p < 0.05$	5.6%	5.6%
Asymptomatic — mean p	0.481	0.490
Asymptomatic — sd p	0.291	0.291

Asymptomatic — % $p < 0.05$	5.7%	5.7%
-----------------------------	------	------

These quantities are Monte Carlo estimates ($n_{\text{perm}} = 10000$ per split) aggregated over dozens of partitions, so variation of this kind is expected relative to the baseline even when reproducing the algorithm exactly — in fact, the main test above (computed with the same seed, in the same order) reproduces the reference p-value exactly, and `vtperm` was validated bit-for-bit against `tperm.fd`. That is why the tolerance here is looser than in the main test.

In both groups, the fraction of splits with $p < 0.05$ (5.6% symptomatic, 5.7% asymptomatic) is consistent with the nominal 5% expected under H_0 , and mean p-values hover around 0.5 — the test does **not** produce systematic false positives when comparing sub-samples of the same group, which calibrates the significance reported for the asymptomatic-symptomatic contrast in the previous section.

Appendix

Validation of the `landscape(..., KK = 1)` reimplementation

```
library(TDA)
tseq <- seq(0, 1, length.out = 500)
maxdiff <- 0
for (f in ph_files) {
  PH <- read.csv(f, header = FALSE)
  colnames(PH) <- c("Birth", "Death", "dimension")
  PHm <- as.matrix(PH[c("dimension", "Birth", "Death")])
  ref_landscape <- landscape(PHm, dimension = 1, KK = 1, tseq)
  mine_landscape <- r_landscape1(PH, dim = 1, tseq = tseq)
  maxdiff <- max(maxdiff, max(abs(ref_landscape - mine_landscape)))
}
maxdiff # 0 across the 17 subjects
```

Validation of `vtperm` against `tperm.fd`

```
set.seed(123)
ref_test <- tperm.fd(fdAsymp, fdSymp, nperm = 500, q = 0.05, plotres = FALSE)
set.seed(123)
mine_test <- vtperm(cbind(eval.fd(argvals, fdAsymp), eval.fd(argvals, fdSymp)),
                    1:8, 9:17, nperm = 500)
isTRUE(all.equal(ref_test$pval, mine_test$pval)) # TRUE
isTRUE(all.equal(ref_test$Tobs, mine_test$Tobs)) # TRUE
```


Full code (reproducible)

```
library(fda)
library(ggplot2)
library(ggrepel)
library(patchwork)
library(igraph)

# Baseline reference values (for comparison only, NOT shown as a result:
# every number reported in the body of the document is recomputed from the
# CSVs committed under output/PH and data/GRN_adjacency_matrices).
ref <- list(
  pc1_pct = 63.5, pc2_pct = 20.0,
  test1_p = 0.0011, test1_Tobs = 6.88, test1_crit95 = 3.72,
  symp_mean = 0.496, symp_sd = 0.290, symp_frac05 = 5.6,
  asymp_mean = 0.481, asymp_sd = 0.291, asymp_frac05 = 5.7
)

# Unified color palette for all figures in the document
col_asymp <- "blue"
col_symp <- "orange"
adj_dir <- "../data/GRN_adjacency_matrices"
adj_files <- list.files(adj_dir, full.names = TRUE)

adj_check <- lapply(adj_files, function(f) as.matrix(read.csv(f, header = FALSE)))
dims_ok <- all(vapply(adj_check, function(m) all(dim(m) == c(250, 250)), logical(1)))
max_asym <- max(vapply(adj_check, function(m) max(abs(m - t(m))), numeric(1)))

stopifnot(
  "Expected 17 adjacency matrices" = length(adj_files) == 17,
  "All matrices must be 250x250" = dims_ok,
  "All matrices must be symmetric (undirected)" = max_asym == 0
)

build_thresholded_graph <- function(adj_path, n_nodes = 60, seed = 0) {
  adj <- as.matrix(read.csv(adj_path, header = FALSE))
  set.seed(seed)
  nodes <- sort(sample(seq_len(nrow(adj)), n_nodes))
  sub <- adj[nodes, nodes]
  thr <- quantile(sub[upper.tri(sub)], 0.85)
  adj_thr <- sub
  adj_thr[adj_thr <= thr] <- 0
  igraph::graph_from_adjacency_matrix(adj_thr, mode = "upper", weighted = TRUE, diag = FALSE)
}

gA <- build_thresholded_graph(file.path(adj_dir, "Asymptomaticadjmatrix2.csv"))
gS <- build_thresholded_graph(file.path(adj_dir, "Symptomaticadjmatrix1.csv"))
```

```

w_rng <- range(c(igraph::E(gA)$weight, igraph::E(gS)$weight))
edge_width <- function(g) 0.4 + 1.6 * (igraph::E(g)$weight - w_rng[1]) / diff(w_rng)

set.seed(1); layoutA <- igraph::layout_in_circle(gA)
set.seed(1); layoutS <- igraph::layout_in_circle(gS)

par(mfrow = c(1, 2), mar = c(1, 1, 3, 1))
plot(gA, layout = layoutA, vertex.size = 5, vertex.label = NA,
     vertex.color = col_asymp, vertex.frame.color = col_asymp,
     edge.width = edge_width(gA), edge.color = grDevices::adjustcolor(col_asymp, alpha.f = 0.4),
     main = "Asymptomatic (subject 2)")
plot(gS, layout = layoutS, vertex.size = 5, vertex.label = NA,
     vertex.color = col_symp, vertex.frame.color = col_symp,
     edge.width = edge_width(gS), edge.color = grDevices::adjustcolor(col_symp, alpha.f = 0.4),
     main = "Symptomatic (subject 1)")

par(mfrow = c(1, 1))
ph_dir <- "../output/PH"
ph_files_raw <- list.files(ph_dir, pattern = "_PH.csv$", full.names = TRUE)

all_dims <- unlist(lapply(ph_files_raw, function(f) {
  d <- read.csv(f, header = FALSE)
  unique(d[[3]])
}))
unique_dims <- sort(unique(all_dims))

stopifnot(
  "Persistence diagrams should only contain dimensions 0 and 1" =
    identical(unique_dims, c(0, 1))
)
tseq <- seq(0, 1, length.out = 500)

# Pure-R reimplementaion of TDA::landscape(..., KK = 1): avoids the heavy
# TDA/GUDHI dependency. Formula (validated bit-for-bit against TDA::landscape,
# see appendix):  $\lambda_1(t) = \max_i \max(0, \min(t - b_i, d_i - t))$  over the
# bars of the requested dimension.
r_landscape1 <- function(PH, dim = 1, tseq) {
  bd <- PH[PH$dimension == dim, c("Birth", "Death"), drop = FALSE]
  if (nrow(bd) == 0) return(rep(0, length(tseq)))
  b <- bd$Birth
  d <- bd$Death
  apply(tseq, function(t) max(0, max(pmin(t - b, d - t))))
}

get_number <- function(fname) as.numeric(sub(".*adjmatrix([0-9]+)_PH.*", "\\1", fname))
ph_files <- ph_files_raw[order(sapply(ph_files_raw, get_number))]

DataSymp <- c()

```

```

DataAsymp <- c()
for (f in ph_files) {
  PH <- read.csv(f, header = FALSE)
  colnames(PH) <- c("Birth", "Death", "dimension")
  x1 <- r_landscapel(PH, dim = 1, tseq = tseq)

  fname <- basename(f)
  if (grepl("^Asymptomatic", fname)) {
    DataAsymp <- cbind(x1, DataAsymp)
  } else if (grepl("^Symptomatic", fname)) {
    DataSymp <- cbind(x1, DataSymp)
  }
}
DataSymp <- t(DataSymp)
DataAsymp <- t(DataAsymp)
DataBoth <- rbind(DataAsymp, DataSymp)

stopifnot(
  "Expected 8 asymptomatic subjects" = nrow(DataAsymp) == 8,
  "Expected 9 symptomatic subjects" = nrow(DataSymp) == 9
)

# B-spline smoothing identical to funTDA.R: order-2 basis (piecewise linear),
# nbasis = Q - 1 = 499 knots over [0, 1].
create_fd <- function(Data) {
  x <- seq(0, 1, length.out = ncol(Data))
  basis <- create.bspline.basis(c(0, 1), nbasis = ncol(Data) - 1, norder = 2)
  fdobj <- smooth.basis(x, t(Data), basis)
  fdobj$fd
}

fdSymp <- create_fd(DataSymp)
fdAsymp <- create_fd(DataAsymp)
fdBoth <- create_fd(DataBoth)
PH_s1 <- read.csv(file.path(ph_dir, "Symptomaticadjmatrix1_PH.csv"), header = FALSE)
colnames(PH_s1) <- c("Birth", "Death", "dimension")

diagram_plot <- ggplot(PH_s1, aes(x = Birth, y = Death, shape = factor(dimension))) +
  geom_abline(slope = 1, intercept = 0, linetype = "dashed", color = "grey50") +
  geom_point(size = 2, alpha = 0.7, color = col_symp) +
  scale_shape_manual(values = c("0" = 16, "1" = 17), name = "Dimension",
    labels = c("H0", "H1")) +
  labs(x = "Birth", y = "Death", title = "Persistence diagram") +
  coord_equal() +
  theme_classic(base_size = 13) +
  theme(plot.title = element_text(hjust = 0.5, face = "bold"))

```

```

ls_s1 <- r_landscape1(PH_s1, dim = 1, tseq = tseq)
landscape_plot <- ggplot(data.frame(t = tseq, lambda = ls_s1), aes(x = t, y = lambda)) +
  geom_line(color = col_symp, linewidth = 0.9) +
  labs(x = "t", y = expression(lambda[1](t)), title = "Persistence landscape (H1)") +
  theme_classic(base_size = 13) +
  theme(plot.title = element_text(hjust = 0.5, face = "bold"))

diagram_plot + landscape_plot
indiv_list <- list()
for (i in seq_len(nrow(DataAsymp))) {
  indiv_list[[length(indiv_list) + 1]] <- data.frame(
    t = tseq, lambda = DataAsymp[i, ], subject = paste0("A", i), Group = "Asymptomatic"
  )
}
for (i in seq_len(nrow(DataSymp))) {
  indiv_list[[length(indiv_list) + 1]] <- data.frame(
    t = tseq, lambda = DataSymp[i, ], subject = paste0("S", i), Group = "Symptomatic"
  )
}
indiv_df <- do.call(rbind, indiv_list)
y_max <- max(indiv_df$lambda) * 1.05

plot_group_landscapes <- function(df, group_label, col) {
  ggplot(df, aes(x = t, y = lambda, group = subject)) +
    geom_line(color = col, alpha = 0.6, linewidth = 0.6) +
    scale_y_continuous(limits = c(0, y_max)) +
    labs(x = "t", y = expression(lambda[1](t)), title = group_label) +
    theme_classic(base_size = 13) +
    theme(plot.title = element_text(hjust = 0.5, face = "bold"))
}

plot_asymp <- plot_group_landscapes(indiv_df[indiv_df$Group == "Asymptomatic", ], "Asymptomatic")
plot_symp <- plot_group_landscapes(indiv_df[indiv_df$Group == "Symptomatic", ], "Symptomatic")

plot_asymp + plot_symp
pca <- pca.fd(fdBoth, nharm = 2)

pc1_pct <- pca$varprop[1] * 100
pc2_pct <- pca$varprop[2] * 100
cum_pct <- pc1_pct + pc2_pct

m <- c("A1", "A2", "A3", "A4", "A5", "A6", "A7", "A8",
      "S1", "S2", "S3", "S4", "S5", "S6", "S7", "S8", "S9")
group <- c(rep("Asymptomatic", 8), rep("Symptomatic", 9))

pca_df <- data.frame(
  PC1 = pca$scores[, 1],

```

```

PC2 = pca$scores[, 2],
ID = m,
Group = group
)

n_asymp_pos <- sum(pca_df$PC1[pca_df$Group == "Asymptomatic"] > 0)
n_symp_neg <- sum(pca_df$PC1[pca_df$Group == "Symptomatic"] < 0)
knitr::kable(
  data.frame(
    Component = c("PC1", "PC2", "Cumulative (PC1+PC2)"),
    `"% variance explained"` = c(sprintf("%.2f%%", pc1_pct), sprintf("%.2f%%", pc2_pct), sprintf(
    check.names = FALSE
  ),
  align = "lr"
)
ggplot(pca_df, aes(x = PC1, y = PC2, color = Group, label = ID)) +
  geom_hline(yintercept = 0, linetype = "dashed", color = "black") +
  geom_vline(xintercept = 0, linetype = "dashed", color = "black") +
  geom_point(size = 4, alpha = 0.9) +
  ggrepel::geom_text_repel(size = 4, show.legend = FALSE, max.overlaps = Inf) +
  scale_color_manual(values = c("Asymptomatic" = col_asymp, "Symptomatic" = col_symp)) +
  labs(x = "PC1 Score", y = "PC2 Score", title = "") +
  coord_fixed() +
  theme_classic(base_size = 16) +
  theme(
    legend.position = "right",
    legend.title = element_blank(),
    plot.title = element_text(hjust = 0.5, face = "bold")
  )
pca_full_df <- data.frame(
  ID = pca_df$ID,
  Group = pca_df$Group,
  PC1 = round(pca_df$PC1, 4),
  PC2 = round(pca_df$PC2, 4)
)
knitr::kable(pca_full_df, align = "llrr")
tol_fpca <- 0.1 # percentage points
stopifnot(
  "PC1 out of tolerance relative to baseline" = abs(pc1_pct - ref$pc1_pct) < tol_fpca,
  "PC2 out of tolerance relative to baseline" = abs(pc2_pct - ref$pc2_pct) < tol_fpca,
  "Expected all 8 asymptomatic subjects with PC1 > 0" = n_asymp_pos == 8,
  "Expected exactly 8 of 9 symptomatic subjects with PC1 < 0" = n_symp_neg == 8
)
rowVar <- function(M) {
  n <- ncol(M)
  mu <- rowMeans(M)
  rowSums((M - mu)^2) / (n - 1)
}

```

```

}

vtperm <- function(X, i1, i2, nperm = 10000) {
  x1 <- X[, i1, drop = FALSE]
  x2 <- X[, i2, drop = FALSE]
  n1 <- ncol(x1); n2 <- ncol(x2)
  Xmat <- cbind(x1, x2)
  n <- n1 + n2
  Ts <- function(M) {
    m1 <- rowMeans(M[, 1:n1, drop = FALSE])
    m2 <- rowMeans(M[, n1 + (1:n2), drop = FALSE])
    v1 <- rowVar(M[, 1:n1, drop = FALSE]) / n1
    v2 <- rowVar(M[, n1 + (1:n2), drop = FALSE]) / n2
    max(abs(m1 - m2) / sqrt(v1 + v2))
  }
  Tobs <- Ts(Xmat)
  Tn <- numeric(nperm)
  for (i in 1:nperm) Tn[i] <- Ts(Xmat[, sample(n)])
  list(pval = mean(Tobs < Tn), Tobs = Tobs, crit95 = unname(quantile(Tn, 0.95)))
}

argvals <- seq(0, 1, length.out = 101)
cache_file <- "../output/results_partA_permtest_cache.rds"

if (file.exists(cache_file)) {
  cached <- readRDS(cache_file)
} else {
  set.seed(123)

  # 1) Main test: Asymptomatic vs Symptomatic
  XFull <- cbind(eval.fd(argvals, fdAsymp), eval.fd(argvals, fdSymp))
  test1 <- vtperm(XFull, 1:8, 9:17, nperm = 10000)

  # 2) Symptomatic split-half: combn(1:9, 4) = 126 4-vs-5 partitions
  XSymp <- eval.fd(argvals, fdSymp)
  splitsS <- combn(1:9, 4)
  pvalsS <- numeric(ncol(splitsS))
  for (i in seq_len(ncol(splitsS))) {
    g1 <- splitsS[, i]; g2 <- setdiff(1:9, g1)
    pvalsS[i] <- vtperm(XSymp, g1, g2, nperm = 10000)$pval
  }

  # 3) Asymptomatic split-half: combn(1:8, 4) = 70 4-vs-4 partitions
  XAsymp <- eval.fd(argvals, fdAsymp)
  splitsA <- combn(1:8, 4)
  pvalsA <- numeric(ncol(splitsA))
  for (i in seq_len(ncol(splitsA))) {
    g1 <- splitsA[, i]; g2 <- setdiff(1:8, g1)
  }
}

```

```

    pvalsA[i] <- vtperm(XAsymp, g1, g2, nperm = 10000)$pval
  }

  cached <- list(test1 = test1, pvalsS = pvalsS, pvalsA = pvalsA)
  saveRDS(cached, cache_file)
}

test1 <- cached$test1
pvalsS <- cached$pvalsS
pvalsA <- cached$pvalsA

stopifnot(
  "Expected 126 symptomatic splits (combn(9,4))" = length(pvalsS) == 126,
  "Expected 70 asymptomatic splits (combn(8,4))" = length(pvalsA) == 70
)
tol_p <- 0.002
tol_T <- 0.05
stopifnot(
  "Main test p-value out of tolerance" = abs(test1$pval - ref$test1_p) < tol_p,
  "Observed Tmax out of tolerance" = abs(test1$Tobs - ref$test1_Tobs) < tol_T,
  "95% critical value out of tolerance" = abs(test1$crit95 - ref$test1_crit95) < tol_T
)
splithalf_summary <- data.frame(
  Group = c("Symptomatic (126 splits, 4 vs 5)", "Asymptomatic (70 splits, 4 vs 4)"),
  `mean p-value` = sprintf("%.3f", c(mean(pvalsS), mean(pvalsA))),
  `sd(p-value)` = sprintf("%.3f", c(sd(pvalsS), sd(pvalsA))),
  `% splits with p < 0.05` = sprintf("%.1f%%", c(mean(pvalsS < 0.05), mean(pvalsA < 0.05)) * 100),
  check.names = FALSE
)
knitr::kable(splithalf_summary, align = "lrrr")
histS <- ggplot(data.frame(p = pvalsS), aes(x = p)) +
  geom_histogram(binwidth = 0.05, boundary = 0, fill = col_symp, alpha = 0.6, color = "white") +
  geom_vline(xintercept = 0.05, linetype = "dashed", color = "black") +
  labs(title = "Symptomatic (126 splits)", x = "p-value", y = "Frequency") +
  theme_classic(base_size = 13)

histA <- ggplot(data.frame(p = pvalsA), aes(x = p)) +
  geom_histogram(binwidth = 0.05, boundary = 0, fill = col_asymp, alpha = 0.6, color = "white") +
  geom_vline(xintercept = 0.05, linetype = "dashed", color = "black") +
  labs(title = "Asymptomatic (70 splits)", x = "p-value", y = "Frequency") +
  theme_classic(base_size = 13)

histA + histS
tol_mean <- 0.02
tol_sd <- 0.02
tol_frac <- 2.0 # percentage points
stopifnot(

```

```

"Symptomatic mean p out of tolerance" = abs(mean(pvalsS) - ref$symp_mean) < tol_mean,
"Symptomatic sd p out of tolerance"   = abs(sd(pvalsS) - ref$symp_sd) < tol_sd,
"Symptomatic frac<0.05 out of tolerance" = abs(mean(pvalsS < 0.05) * 100 - ref$symp_frac05) < tol_frac05,
"Asymptomatic mean p out of tolerance" = abs(mean(pvalsA) - ref$asymp_mean) < tol_mean,
"Asymptomatic sd p out of tolerance"   = abs(sd(pvalsA) - ref$asymp_sd) < tol_sd,
"Asymptomatic frac<0.05 out of tolerance" = abs(mean(pvalsA < 0.05) * 100 - ref$asymp_frac05) < tol_frac05,
)
library(TDA)
tseq <- seq(0, 1, length.out = 500)
maxdiff <- 0
for (f in ph_files) {
  PH <- read.csv(f, header = FALSE)
  colnames(PH) <- c("Birth", "Death", "dimension")
  PHm <- as.matrix(PH[c("dimension", "Birth", "Death")])
  ref_landscape <- landscape(PHm, dimension = 1, KK = 1, tseq)
  mine_landscape <- r_landscape1(PH, dim = 1, tseq = tseq)
  maxdiff <- max(maxdiff, max(abs(ref_landscape - mine_landscape)))
}
maxdiff # 0 across the 17 subjects
set.seed(123)
ref_test <- tperm.fd(fdAsymp, fdSymp, nperm = 500, q = 0.05, plotres = FALSE)
set.seed(123)
mine_test <- vtperm(cbind(eval.fd(argvals, fdAsymp), eval.fd(argvals, fdSymp)),
  1:8, 9:17, nperm = 500)
isTRUE(all.equal(ref_test$pval, mine_test$pval)) # TRUE
isTRUE(all.equal(ref_test$Tobs, mine_test$Tobs)) # TRUE
sessionInfo()

```

Session information

```
sessionInfo()
```

R version 4.5.3 (2026-03-11)

Platform: aarch64-apple-darwin20

Running under: macOS Tahoe 26.5.2

Matrix products: default

BLAS: /Library/Frameworks/R.framework/Versions/4.5-arm64/Resources/lib/libRblas.0.dylib

LAPACK: /Library/Frameworks/R.framework/Versions/4.5-arm64/Resources/lib/libRlapack.dylib; LAPACK: /Library/Frameworks/R.framework/Versions/4.5-arm64/Resources/lib/libRlapack.dylib

locale:

[1] C.UTF-8/C.UTF-8/C.UTF-8/C/C.UTF-8/C.UTF-8

time zone: Europe/Madrid

tzcode source: internal

attached base packages:

```
[1] splines    stats      graphics  grDevices  utils      datasets  methods
[8] base
```

other attached packages:

```
[1] igraph_2.3.2    patchwork_1.3.2 ggrepel_0.9.8    ggplot2_4.0.3
[5] fda_6.3.0       deSolve_1.42    fds_1.9          RCurl_1.98-1.19
[9] rainbow_3.8     pcaPP_2.0-5     MASS_7.3-65
```

loaded via a namespace (and not attached):

```
[1] ks_1.15.2        generics_0.1.4    bitops_1.0-9     KernSmooth_2.23-26
[5] lattice_0.22-9   hdrdce_3.5.0      pracma_2.4.6     digest_0.6.39
[9] magrittr_2.0.5   evaluate_1.0.5    grid_4.5.3       RColorBrewer_1.1-3
[13] mvtnorm_1.4-2    fastmap_1.2.0     jsonlite_2.0.0   Matrix_1.7-4
[17] mclust_6.1.3     scales_1.4.0      cli_3.6.6        rlang_1.2.0
[21] withr_3.0.2      yaml_2.3.12       tools_4.5.3      dplyr_1.2.1
[25] colorspace_2.1-3 vctrs_0.7.3       R6_2.6.1         lifecycle_1.0.5
[29] cluster_2.1.8.2  pkgconfig_2.0.3   pillar_1.11.1    gtable_0.3.6
[33] glue_1.8.1       Rcpp_1.1.1        xfun_0.57        tibble_3.3.1
[37] tidyselect_1.2.1 knitr_1.51        farver_2.1.2     htmltools_0.5.9
[41] rmarkdown_2.31   labeling_0.4.3    compiler_4.5.3   S7_0.2.2
```